



Humans Can Rapidly Learn What to Encode in Memory

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Summary

- Learning what to encode is a long-range credit-assignment problem: The utility of encoding a memory is often unclear until later.
- We designed a behavioral task to manipulate demands for selective encoding.
- Pilot results suggest that humans can learn to selectively encode task-relevant information.
- A neural network that uses retrieved episodic memories to support encoding policy updates can qualitatively reproduce the human data.

Retrieval may bridge the long encoding–outcome delay

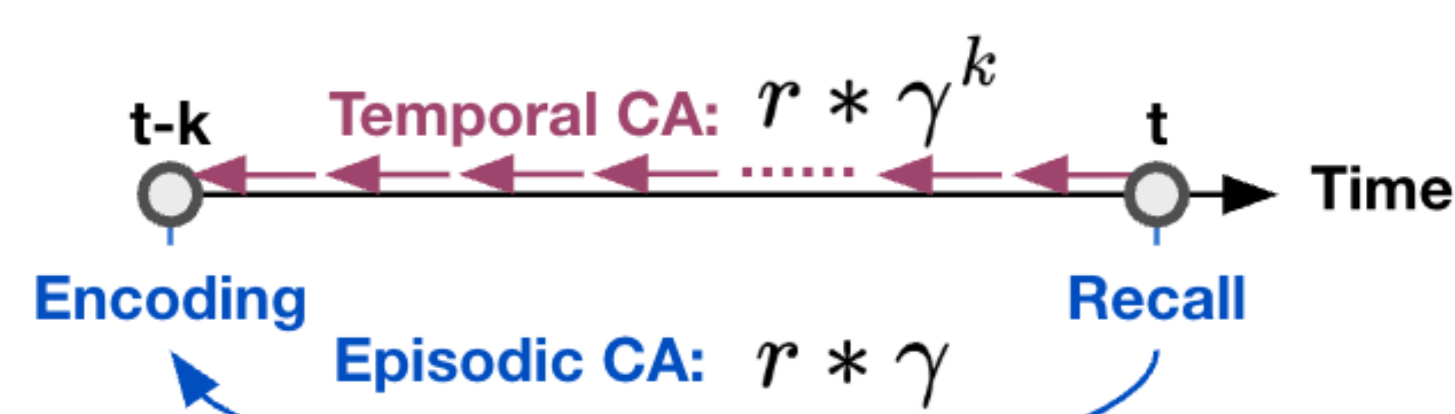


Figure 1: Temporal versus episodic credit assignment (CA).

- Temporal credit decays exponentially over time, making encoding policy – what to encode in memory – difficult to learn (Sutton & Barto, 2018).
- For reward r , temporal CA discounts an encoding action k steps in the past by $r\gamma^k$, where $0 < \gamma < 1$ is the discount factor (1).
- Episodic retrieval can bridge the delay to support long-range learning (Gershman & Daw, 2017) by acting as a "skip connection" to credit distant encoding actions that eventually led to the reward.

References

[1] Gershman & Daw (2017). Reinforcement learning and episodic memory in humans and animals: An integrative framework. *Annual Review of Psychology*, 68(1), 101–128. [2] Sutton R.S. & Barto A.G. (2018). *Reinforcement learning: An introduction*. MIT Press. [3] Stark, S. M., Kirwan, C. B., & Stark, C. E. L. (2019). Mnemonic Similarity Task: A tool for assessing hippocampal integrity. *Trends in Cognitive Sciences*, 23(11), 938–951.

A behavioral paradigm to induce selective encoding

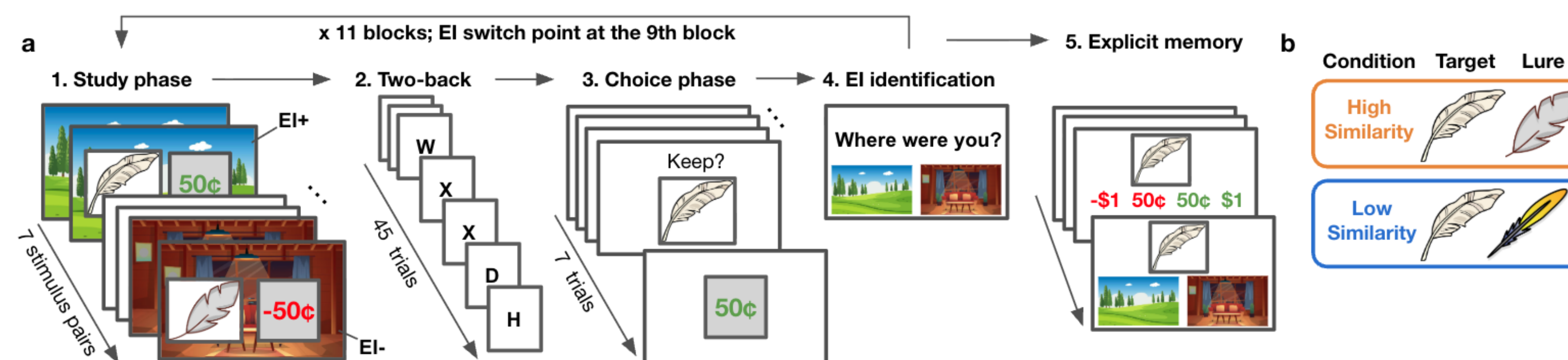


Figure 2: The behavioral paradigm. EI = encoding indicator; EI+ = relevant scene.

- **a**, Participants studied item-value associations in two scenes, completed a two-back task to flush their working memory, and then made value-based choices with the goal of keeping positive-value items and rejecting negative-value items. Only items from one of the scenes (EI+) reappeared at choice. Participants were prompted to identify the EI+ after each block. After 11 blocks, they completed explicit item-scene and item-value memory tests.
- At block 9, the EI+ and EI- were exchanged; selective encoding of EI+ items should lead to impaired performance at reversal—a **switch cost**.
- **b**, Each EI+ target was paired with a random-value EI- lure; target-lure similarity was either high or low (Stark et al., 2019). We predicted a larger switch cost in the high-similarity condition: Participants should be more motivated to selectively encode in this condition, due to a higher potential for interference.

Signature of selective encoding

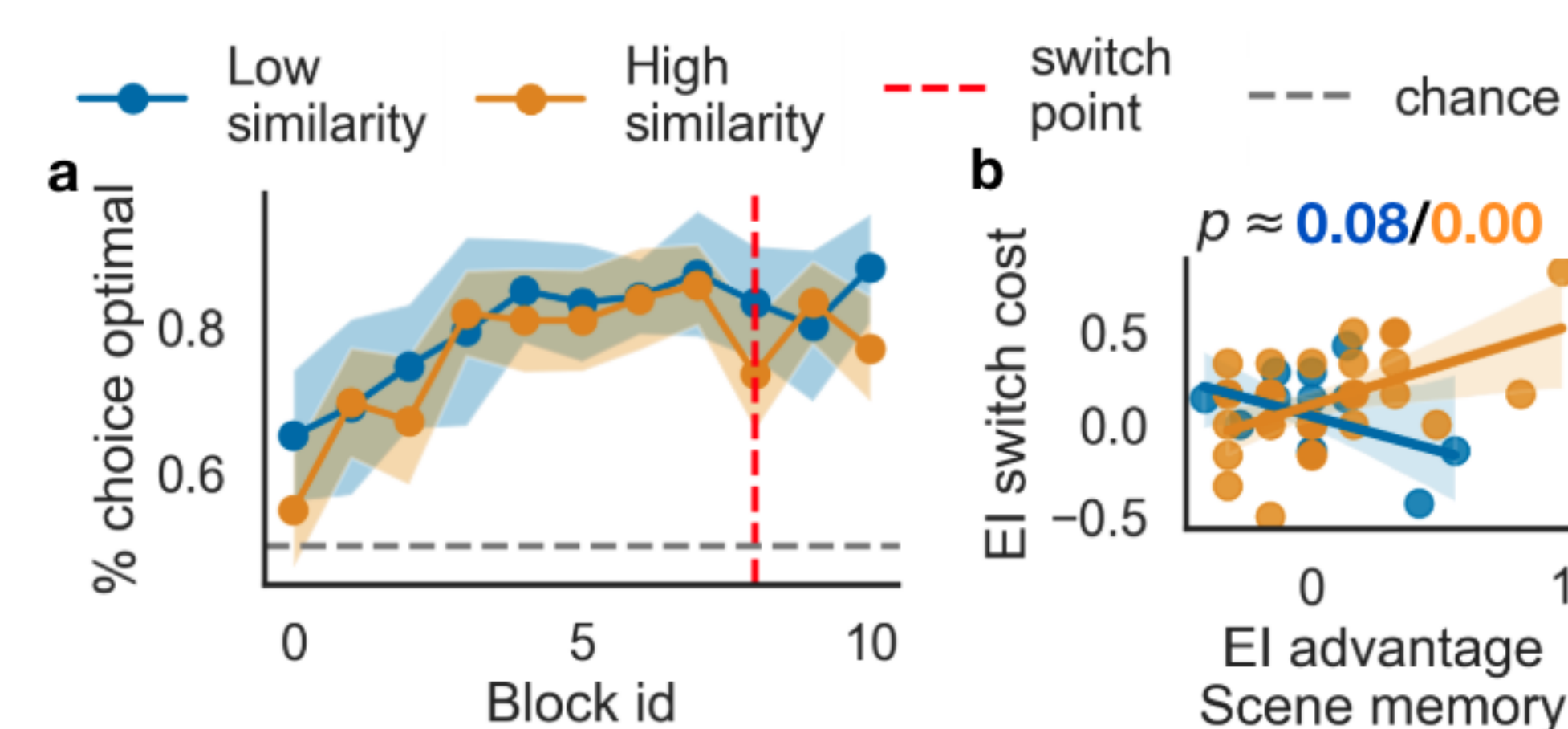


Figure 3: Humans can learn to selectively encode relevant information in the high similarity condition.

- **a**, Participants learned to make value-based choices in both conditions (low similarity, $n = 20$; high similarity, $n = 39$). EI reversal impaired accuracy, with a larger **switch cost** under high similarity condition.
- **b**, In the high similarity condition, participants with greater EI advantage in scene memory had a larger switch cost; this effect was weaker in the low similarity condition.

Selective encoding influences subsequent memory

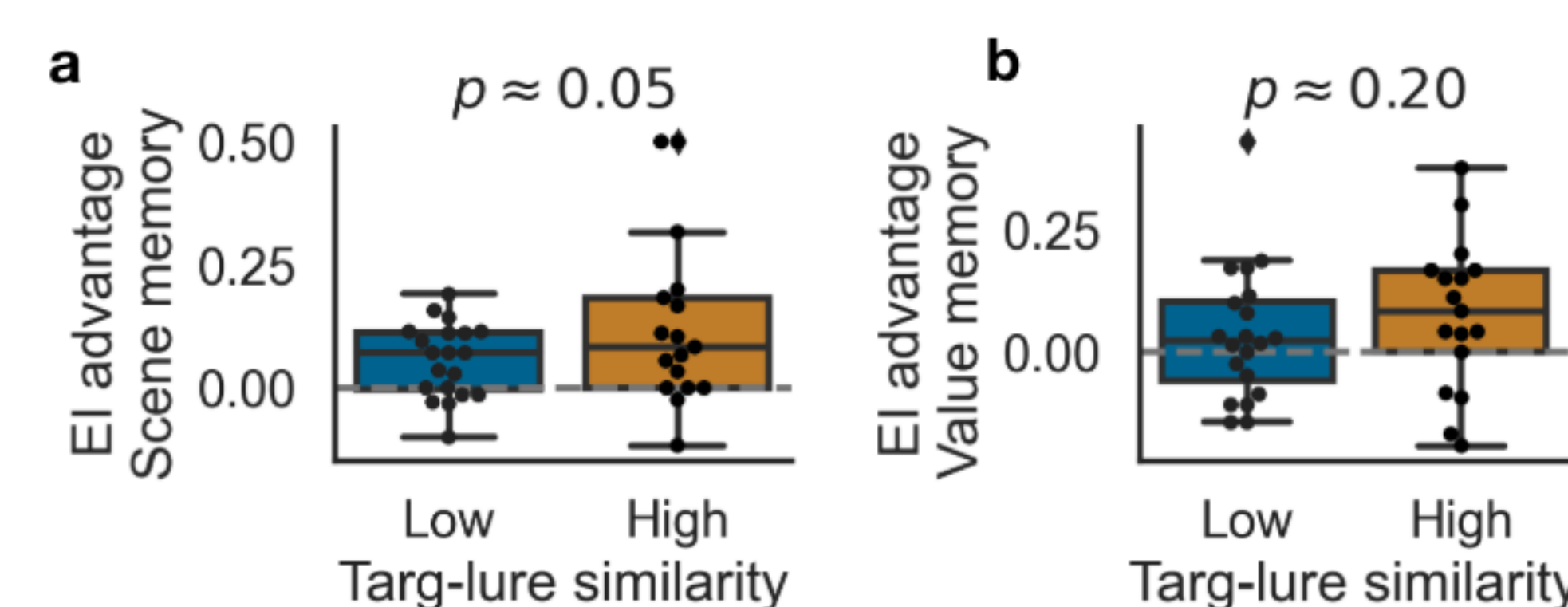


Figure 4: The EI+ memory advantage is stronger when the demand for selective encoding is high.

- **a**. High target–lure similarity numerically increased EI advantage (EI+ minus EI- accuracy) for scene memory ($p \approx 0.05$).
- **b**. The same analysis was done for value memory, but note that EI+ items (but not EI- items) were also present during the choice phase, so EI advantage for value memory cannot be fairly computed.

A neural network model with episodic memory

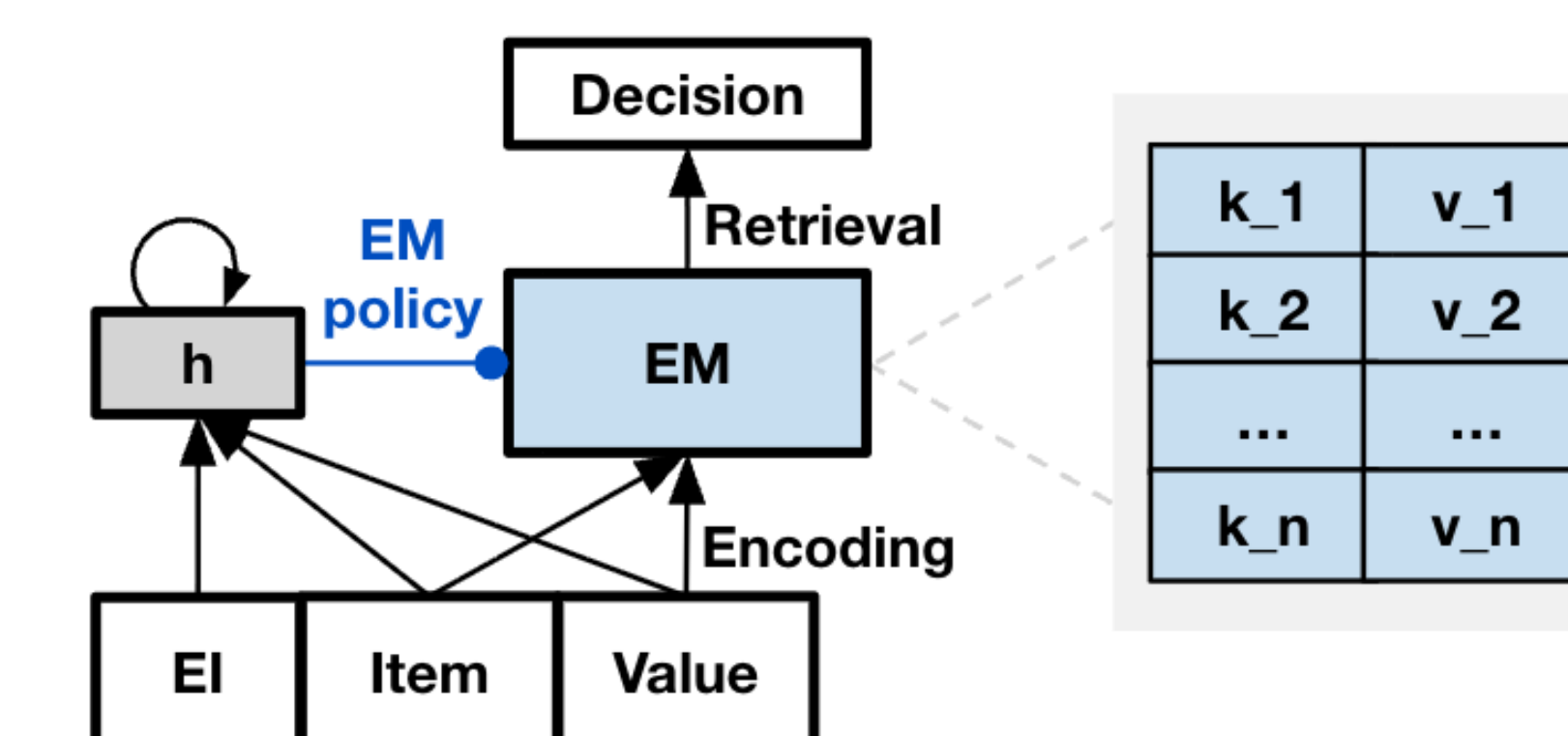


Figure 5: The model architecture.

- A recurrent state h receives the EI, item and value on every trial during study.
- An encoding policy, which decides whether to encode the studied item, gates storage in episodic memory, where items serve as memory keys supporting content-based retrieval of associated values to inform choice.
- In the high similarity condition, target-lure confusion is more likely compared to the low similarity condition by design.
- Retrieval assigns choice outcomes directly to the encoding actions that stored the retrieved memories, allowing the model to learn what encoding actions led to reward eventually.

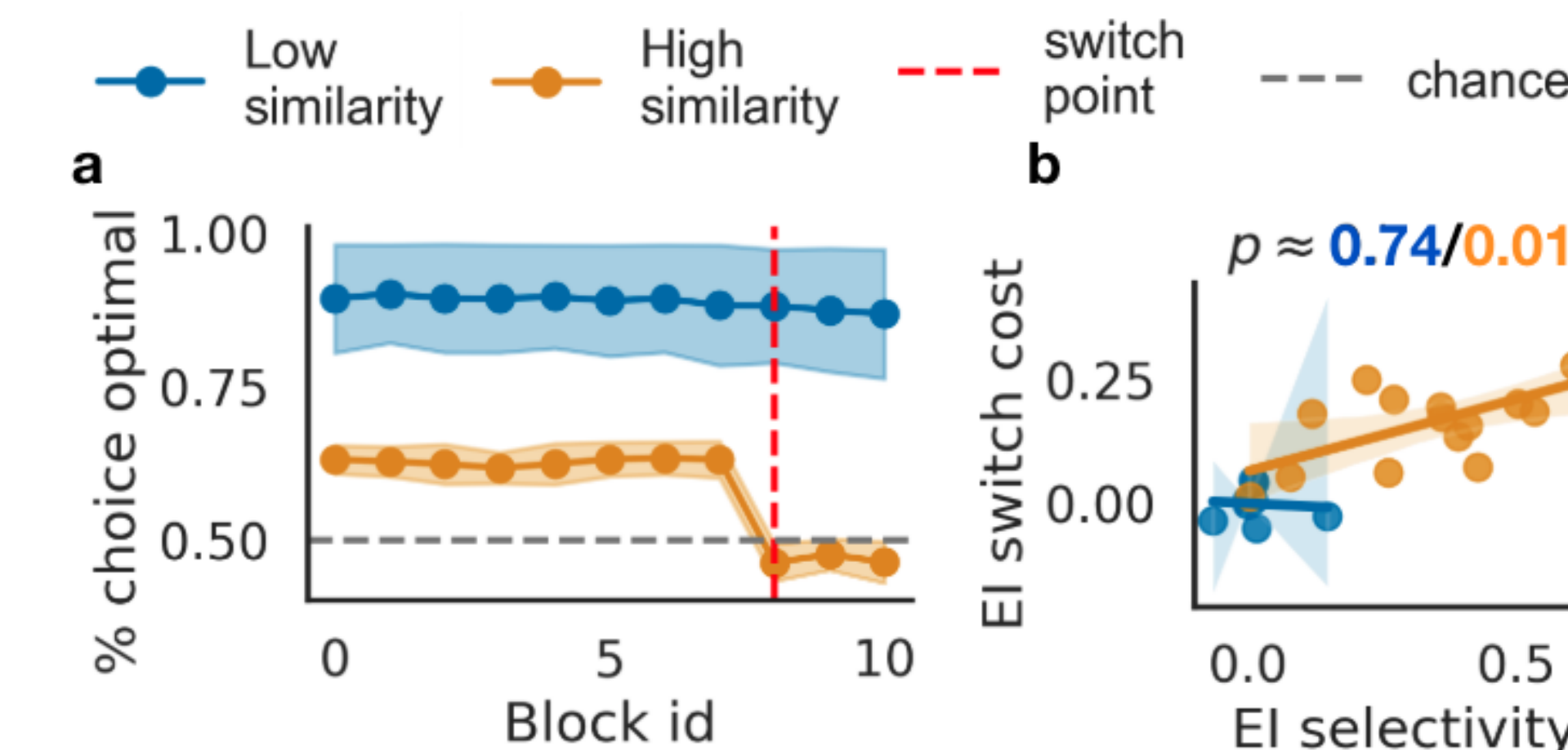


Figure 6: Selective encoding in the model.

- **a**, EI reversal impaired models in the high-similarity condition, which learned to selectively encode EI+ items.
- **b**, EI selectivity (EI+ minus EI- encoding probability) is correlated with switch cost under high similarity ($p \approx 0.01$), but not under low similarity.

Question?

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